

A case study: numerical verification for the disproof of McKean's conjecture on entropy production

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Abstract

In the recent work [6], L. Silvestre disproves McKean's 1966 conjecture [4] that the entropy production of the spatially homogeneous Boltzmann equation decreases monotonically, for the kernel $B = |v - v_*|^\gamma / (4\pi)$ with $\gamma \in [0, 1]$, covering hard-sphere and the Maxwell molecule model. The counterexample is a radial mixture of two Maxwellians $f_R = (1 - p)M_1 + pM_R$, and the theorem is asymptotic as $R \rightarrow \infty$. This report numerically verifies it by three independent routes for the constructed state f_R at finite R by 1D radial algorithm, 1D measure-valued algorithm, and our 3D Fourier spectral solver [2], and further explores the time evolution of entropy production $\partial_t D(f)$ evolving from f_R .

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1 The Problem

The space-homogeneous Boltzmann equation in three dimensions reads

$$\partial_t f = Q(f, f)$$

where

$$Q(f, g)(v) = \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} B(f(v'_*)g(v') - f(v_*)g(v)) \, d\sigma \, dv_*$$

and

$$v' = \frac{v+v_*}{2} + \frac{|v-v_*|}{2}\sigma, \quad v'_* = \frac{v+v_*}{2} - \frac{|v-v_*|}{2}\sigma,$$

and Silvestre [6] works throughout with the kernel

$$B = \frac{1}{4\pi} |v - v_*|^\gamma, \quad \gamma \in [0, 1], \quad (1)$$

so $\gamma = 1$ is hard-sphere and $\gamma = 0$ is Maxwell molecules with a constant angular cross section. It is well-known that

$$H(f) = - \int f \log f \, dv, \quad D(f) = \partial_t H(f) = - \int Q(f, f) \log f \, dv \geq 0,$$

and McKean [4] proposed a conjecture in 1966 that the entropy production $D(f)$ is monotone decreasing in time:

$$\partial_t D \leq 0, \quad \text{i.e.} \quad H'' \leq 0.$$

Recently, Silvestre disproves it in [6, Theorem 1].¹

1.1 Key results

The key results are obtained by constructing the radial mixture of two Maxwellians

$$f_R = (1-p)M_1 + pM_R, \quad M_T(v) = (2\pi T)^{-3/2} e^{-|v|^2/(2T)}, \quad R \geq 2,$$

the [6, Theorem 1] is asymptotic as $R \rightarrow \infty$, i.e., for fixed $p \in (0, p_\gamma)$,

$$\partial_t D(f_R) = c_{p,\gamma} R^\gamma \log R + o_p(R^\gamma \log R), \quad (2)$$

or

$$\lim_{R \rightarrow \infty} \frac{\partial_t D(f_R)}{R^\gamma \log R} = c_{p,\gamma},$$

where

$$c_{p,\gamma} = \frac{3}{2}p(1-p) \left[(1-p) \frac{2-\gamma}{2+\gamma} \kappa_{2\gamma} - p \kappa_\gamma^2 \right], \quad p_\gamma = \left[1 + \frac{(2+\gamma)\kappa_\gamma^2}{(2-\gamma)\kappa_{2\gamma}} \right]^{-1}, \quad (3)$$

with

$$\kappa_s = \int |z|^s M_1 \, dz = 2^{s/2} \Gamma(\frac{3+s}{2}) / \Gamma(\frac{3}{2}).$$

At the endpoints,

$$\gamma = 0: \quad p_0 = \frac{1}{2}, \quad c_{p,0} = \frac{3}{2}p(1-p)(1-2p), \quad (4)$$

$$\gamma = 1: \quad p_1 = \frac{\pi}{\pi+8} \approx 0.282, \quad c_{p,1} = \frac{3}{2}p(1-p) \left(1 - p - \frac{8}{\pi}p \right). \quad (5)$$

So $\partial_t D(f_R) > 0$ for every sufficiently large R whenever $p < p_\gamma$, and H'' changes sign.

¹Recently similar progress for Landau-Coulomb equation and Kac equation are referred to [3] and [8].

1.2 Mechanism

The starting point, inherited from the previous paper [7], is the identity

$$\partial_t D(f) = - \int_{\mathbb{R}^3} \frac{Q^2}{f} dv + \int_{\mathbb{R}^3} Q \mathcal{L} dv, \quad (6)$$

with

$$\mathcal{L}(v) = \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_* \log \frac{ff_*}{f'f'_*} B d\sigma dv_*.$$

This is what produces (2): [6] shows that the first term of (6) is only $O_p(R^\gamma)$ ([6, Lemma 3]) while the second term is $O_p(R^\gamma \log R)$ ([6, Lemma 7]), so the $\log R$ decides the sign. The subtlety that makes the construction hard is described in [6, Page 3]: the integrands of Q and $\mathcal{L}$ have *pointwise opposite signs*, so $\int Q \mathcal{L}$ is generically negative. The essence of the construction is to make them agree on enough of velocity space, by a margin of order $\log R$.

Both terms of (6) are computable, in particular for Gaussian mixtures such as f_R , e.g., writing $f_R = pM_R(1 + \frac{1-p}{p}M_1/M_R)$ gives the exact formula

$$\log f_R(v) = \log \frac{p}{(2\pi R)^{3/2}} - \frac{|v|^2}{2R} + \psi_R(v), \quad \psi_R(v) = \log \left(1 + \frac{1-p}{p} R^{3/2} e^{-\theta_R |v|^2} \right), \quad (7)$$

$\theta_R = \frac{R-1}{2R} \in [\frac{1}{4}, \frac{1}{2}]$. In the integrand of $\mathcal{L}$, the constant and $|v|^2/(2R)$ cancels by conservation of mass and energy, so everything reduces to ψ_R :

$$\log \frac{f_R f_{R*}}{f'_R f'_{R*}} = -\Delta \psi_R, \quad \Delta \psi_R := \psi'_R + \psi'_{R*} - \psi_R - \psi_{R*} \quad (\text{outgoing minus incoming}).$$

To make it clear, we plot ψ_R in Figure 1, which is a somewhat step function of height $\log \frac{1-p}{p} + \frac{3}{2} \log R$ and fall to 0 on the "hot" scale $|v| \sim \sqrt{R}$. A "cold-hot" collision has exactly one "cold" input and, by [6, Eq.(12)], two "hot" outputs, so $\Delta \psi_R$ picks up the full step and nothing cancels. That is the dichotomy $-\Delta \psi_R / \log R \rightarrow \frac{3}{2}$ which drives the result.

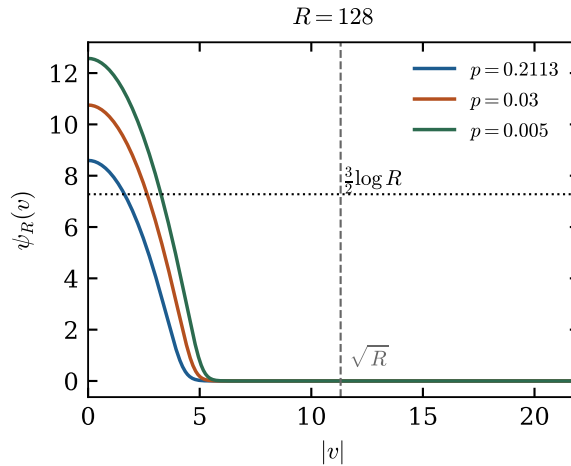


Figure 1: Plot of ψ_R in (7) for $R = 128$. Its height is $\frac{3}{2} \log R + \log \frac{1-p}{p}$: lowering p raises the step just as effectively as raising R . Only the $\frac{3}{2} \log R$ part survives into the asymptotic constant $c_{p,\gamma}$, which is why the p -dependence of the finite- R onset (§2) is invisible in the theorem.

Alternative but equivalent formula of (6). For the sake of numerical verification, we now compute the time derivative of the entropy D , i.e., differentiating D along $\partial_t f = Q(f, f)$ instead gives the natural form

$$\partial_t D = - \int \frac{Q^2}{f} dv - 2 \int Q_{\text{sym}}(f, Q) \log f dv, \quad (8)$$

with $Q_{\text{sym}}(g, h) = \frac{1}{2}[Q(g, h) + Q(h, g)]$ the polarisation of the quadratic form: $\partial_t Q(f, f) = Q(Q, f) + Q(f, Q) = 2Q_{\text{sym}}(f, Q)$ and $\partial_t \log f = Q/f$, so (8) is just the product rule applied to $D = - \int Q(f, f) \log f$.

That (6) and (8) agree is not obvious, and it is the whole reason Tier C is possible, so it is worth doing once. The second term of (8) is *bilinear* in (f, Q) ; the second term of (6) is *linear* in Q , tested against a weight built from f alone. Everything follows from the weak form of the collision operator.

$$\int Q_{\text{sym}}(g, h) \varphi dv = \frac{1}{4} \iiint (gh_* + hg_*) (\varphi' + \varphi'_* - \varphi - \varphi_*) B d\sigma dv_* dv. \quad (9)$$

Now take $g = f$, $h = Q$ and $\varphi = \log f$, so that $\varphi' + \varphi'_* - \varphi - \varphi_* = -\log \frac{ff_*}{f'f'_*}$:

$$-2 \int Q_{\text{sym}}(f, Q) \log f dv = \frac{1}{2} \iiint (fQ_* + Qf_*) \log \frac{ff_*}{f'f'_*} B d\sigma dv_* dv.$$

The logarithm, B and the measure are all symmetric under $v \leftrightarrow v_*$, so relabelling turns the fQ_* term into the Qf_* term: the two contribute equally, the $\frac{1}{2}$ cancels, and the $dv_* d\sigma$ integration collects into a function of v alone,

$$-2 \int Q_{\text{sym}}(f, Q) \log f dv = \int Q(v) \underbrace{\left[\int_{\mathbb{R}^3 \times \mathbb{S}^2} f_* \log \frac{ff_*}{f'f'_*} B d\sigma dv_* \right]}_{= \mathcal{L}(v)} dv,$$

which is (6). So the content of the identity is that a bilinear expression in (f, Q) collapses to Q against a weight depending on f only.

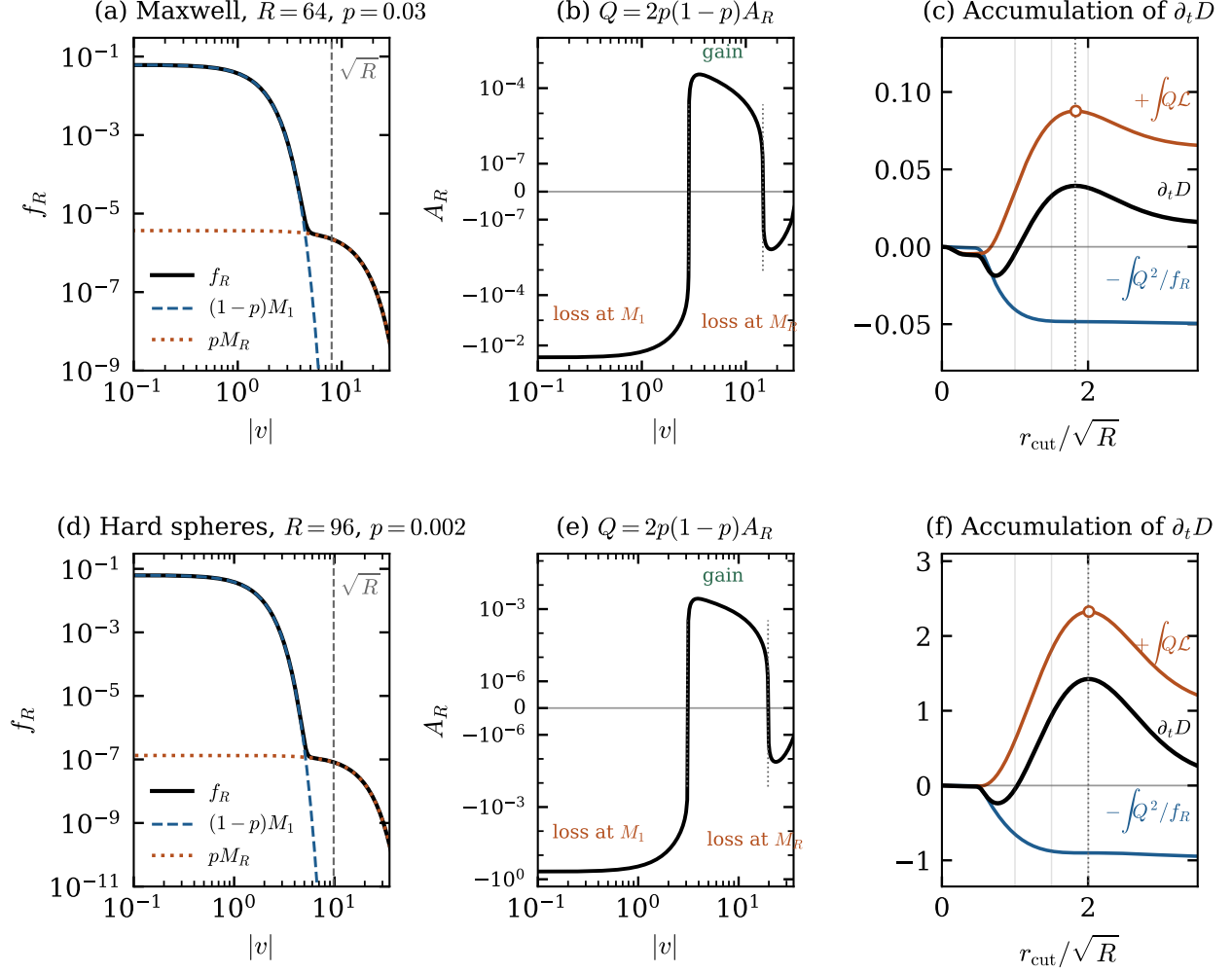


Figure 2: The objects, radially, at each kernel's Tier C production case: $\gamma = 0$, $(p, R) = (0.03, 64)$ on top, $\gamma = 1$, $(p, R) = (0.002, 96)$ below.

1.3 Organization

The following report is a comprehensive verification of the entropy dissipation result using three independent approaches, each supplying something the others cannot; see Table 1 for a summary.

Tier	Representation	Kernel	Conclusion supported
A	1D radial solver for initial state f_R	Maxwell Molecules and Hard-sphere	Check $\partial_t D(f_R)$ to 7–8 digits at any R (asymptotic sense)
B	1D measure-valued solver for time evolution	Maxwell Molecules	Check time-evolution $D(f(t))$ and the duration of the positive- $\partial_t D$ interval for some R (non-asymptotic sense)
C	3D Fourier spectral solver on Cartesian grid	Maxwell Molecules	Check time-evolution $D(f(t))$ and the duration of the positive- $\partial_t D$ interval for some R with no radial symmetry assumed (non-asymptotic sense)

Table 1: Organization of the verification.

2 Tier A Test: asymptotic verification of $\partial_t D(f_R)$ by 1D radial algorithm (Maxwell molecules and Hard sphere)

In this section, we implement Tier A test, we evaluate the static functional (6) directly: every object in it is radial, so the identity collapses to 1D quadratures and $\partial_t D(f_R)$ follows semi-analytically at any R , p and γ . It will also serve as the reference value as well as the only tier that reaches the asymptotic regime.

2.1 Method

Note that, since substitution $\partial_t f = Q(f, f)$ has already been made in the alternative form (8), what remains on the right of (8) is a nonlinear functional of f alone, with no time derivative, no trajectory, nothing to integrate. Evaluating $\partial_t D(f_R)$ means evaluating a static functional at one input function. f_R , A_R and $\mathcal{L}$ are all exactly radial — which is what lets Tier A reduce (6) or (8) to 1D quadratures — so Figure 2 loses nothing.

Every object in (6) evaluated on f_R is radial: f_R , A_R , ψ_R and $\mathcal{L}$. Fix $|v| = r$ and parametrize v_* by $s = |v_*|$ and $\mu = \hat{v} \cdot \hat{v}_*$, and σ by $u = (\widehat{v + v_*}) \cdot \sigma$. Then $dv_* \rightarrow 4\pi s^2 ds$ with μ and u each uniform on $[-1, 1]$ with density $\frac{1}{2}$, and

$$|v - v_*|^2 = r^2 + s^2 - 2rs\mu, \quad |v'|^2, |v'_*|^2 = \frac{r^2 + s^2}{2} \pm \frac{\sqrt{(r^2 + s^2)^2 - 4r^2 s^2 \mu^2}}{2} u, \quad (10)$$

the two signs manifestly conserving energy. This turns $\mathcal{L}$ into a 3D quadrature,

$$\mathcal{L}(r) = 4\pi \int_0^\infty s^2 f_R(s) \left\langle (\psi_R(r) + \psi_R(s) - \psi_R(|v'|) - \psi_R(|v'_*|)) |v - v_*|^\gamma \right\rangle_{\mu, u} ds, \quad (11)$$

and the two terms of (6) into 1D radial integrals, so that

$$\partial_t D(f_R) = \underbrace{-4p^2(1-p)^2 \int \frac{A_R^2}{f_R} dv}_{\text{negative term}} + \underbrace{2p(1-p) \int A_R \mathcal{L} dv}_{\text{no definite sign}}. \quad (12)$$

For $\gamma = 0$ the kernel integrates to $\int_{\mathbb{S}^2} B d\sigma = 1$, so the loss part of $Q(f, g)$ is just g , and the gain part is the $P(f, g)$ in [6, Eq.(22)]. [6, Lemma 4] gives it in closed form,

$$P(M_a, M_b) = \frac{1}{|b - a|} \int_{\min(a, b)}^{\max(a, b)} M_u du, \quad (13)$$

whence, exactly,

$$A_R = P(M_1, M_R) - \frac{1}{2}(M_1 + M_R) \quad (\gamma = 0). \quad (14)$$

For $\gamma > 0$, the theoreical proof in [6] needs only an upper bound, but a numerical verification needs the value. So we extend [6, Lemma 4] to $\gamma > 0$ in the following proposition, which is the only nontrivial step in Tier A. The proof is a straightforward extension of the original.

Proposition 1 (γ -extended Lemma 4 of [6]). *Let $G_{a,b}^{(\gamma)}$ denote the gain part of $\frac{1}{2}[Q(M_a, M_b) + Q(M_b, M_a)]$ for the collision kernel in (1), normalized so that $\int_{\mathbb{S}^2} B d\sigma = |v - v_*|^\gamma$. Then*

$$G_{a,b}^{(\gamma)} = \frac{2^\gamma}{2} \int_{-1}^1 c(t)^{-\gamma} \left[M_\alpha * (|\cdot|^\gamma M_{c(t)^{2\beta}}) \right] dt, \quad c(t)^2 = 1 + \lambda^2 + 2\lambda t, \quad (15)$$

with $\alpha = \frac{ab}{a+b}$, $\beta = \frac{a+b}{4}$, $\lambda = \frac{a-b}{a+b}$.

The proof is a straightforward extension of the $\gamma = 0$ case but should carry with it the additional factor of $(2|z|)^\gamma$. The reason this survives is that the extra factor is $(2|z|)^\gamma$ in the half-difference variable $z = \frac{v-v_*}{2}$, hence *radial in z* : after the substitution $m = \frac{v+v_*}{2}$, $z = \frac{v-v_*}{2}$ and the completion of the square $8M_a(v)M_b(v_*) = M_\alpha(m - \lambda z)M_\beta(z)$, the measure $|z|^\gamma M_\beta(z) dz$ is still radial, so the rotational-averaging step of the original proof may be inserted unchanged — it replaces $\lambda\hat{z} + \sigma$ by $c(t)\eta$ with η uniform and $t = \hat{z} \cdot \sigma$ carrying the measure $\frac{1}{2} dt$. Substituting $\zeta = c(t)z$ then gives the stated form. One can check that setting $\gamma = 0$ degenerates it to

$$\frac{1}{2} \int_{-1}^1 M_{\theta(t)} dt \quad \text{with} \quad \theta(t) = \alpha + c(t)^2 \beta = \frac{a+b}{2} + t \frac{a-b}{2}$$

which is exactly (13) or [6, Lemma 4].

The formula (15) is only a 2D quadrature, because the radial 3D convolution has an elementary inner integral when the Gaussian is placed inside; the same identity gives the collision frequency $\nu_T = |\cdot|^\gamma * M_T$ as a single 1D integral.

2.2 Numerical results

Tables 2 and 3 display (12) term by term, corresponding to [6, Sections 3 and 4] side by side. Reading down the columns, the negative column converges to a constant at $\gamma = 0$ ([6, Lemma 3] with $R^\gamma = 1$) and grows like R at $\gamma = 1$, while the positive column grows like $R^\gamma \log R$ as shown in [6, Lemma 7]. The residual $b_{\text{eff}} := \partial_t D / R^\gamma - c_{p,\gamma} \log R$ flattens rather than diverging, confirming the form $\partial_t D(f_R) = R^\gamma (c_{p,\gamma} \log R + b + o(1))$ as shown in Figure 3.

R	$-4p^2(1-p)^2 \int A_R^2 / f_R$	$2p(1-p) \int A_R \mathcal{L}$	$\partial_t D(f_R)$	$c_{p,0} \log R$	b_{eff}
$p = 0.2113$ (asymptotic argmax), $c_{p,0} = 0.144338$					
8	-0.038616	-0.028647	-0.067263	0.30014	-0.3674
32	-0.186647	+0.035305	-0.151342	0.50024	-0.6516
128	-0.424501	+0.287651	-0.136851	0.70033	-0.8372
512	-0.659067	+0.627728	-0.031339	0.90042	-0.9318
1024	-0.755638	+0.794053	+0.038415	1.00047	-0.9621
4096	-0.899357	+1.091090	+0.191736	1.20057	-1.0088
$p = 0.03$, $c_{p,0} = 0.041031$					
8	-0.003505	-0.006159	-0.009664	0.08532	-0.0950
32	-0.027057	+0.023831	-0.003226	0.14220	-0.1454
64	-0.049576	+0.064432	+0.014856	0.17064	-0.1558
128	-0.076047	+0.116376	+0.040330	0.19908	-0.1588
1024	-0.051189 [†]	+0.107356 [†]	+0.056167[†]	0.10087 [†]	-0.0447 [†]

Table 2: Maxwell molecules, $\gamma = 0$. Bold entries are positive, i.e. McKean’s conjecture fails. [†]this row is from the $p = 0.01$ block, included to show the small- p behavior.

R	$-4p^2(1-p)^2 \int A_R^2/f_R$	$2p(1-p) \int A_R \mathcal{L}$	$\partial_t D(f_R)$	$\partial_t D/R$	b_{eff}
$p = 0.1296$ (asymptotic argmax), $c_{p,1} = 0.091435$					
64	-34.174	+8.273	-25.901	-0.404705	-0.7850
512	-439.556	+257.522	-182.033	-0.355534	-0.9259
4096	-4147.83	+3128.33	-1019.51	-0.248903	-1.0094
$p = 0.03$, $c_{p,1} = 0.039006$					
128	-22.3659	+19.2448	-3.12105	-0.024383	-0.2136
256	-53.1911	+53.0072	-0.18385	-0.000718	-0.2170
512	-119.841	+130.994	+11.1532	+0.021784	-0.2215
4096	-1151.57	+1504.36	+352.787	+0.086130	-0.2383
$p = 0.01$, $c_{p,1} = 0.014323$					
32	-0.908586	+0.256627	-0.651960	-0.020374	-0.0700
64	-2.80202	+2.19051	-0.611505	-0.009555	-0.0691
128	-7.43179	+7.71804	+0.286243	+0.002236	-0.0673
512	-40.7812	+52.5426	+11.7614	+0.022971	-0.0664
4096	-396.188	+594.805	+198.617	+0.048490	-0.0706

Table 3: Hard spheres, $\gamma = 1$, using the γ -extended Proposition 1. Bold entries are positive.

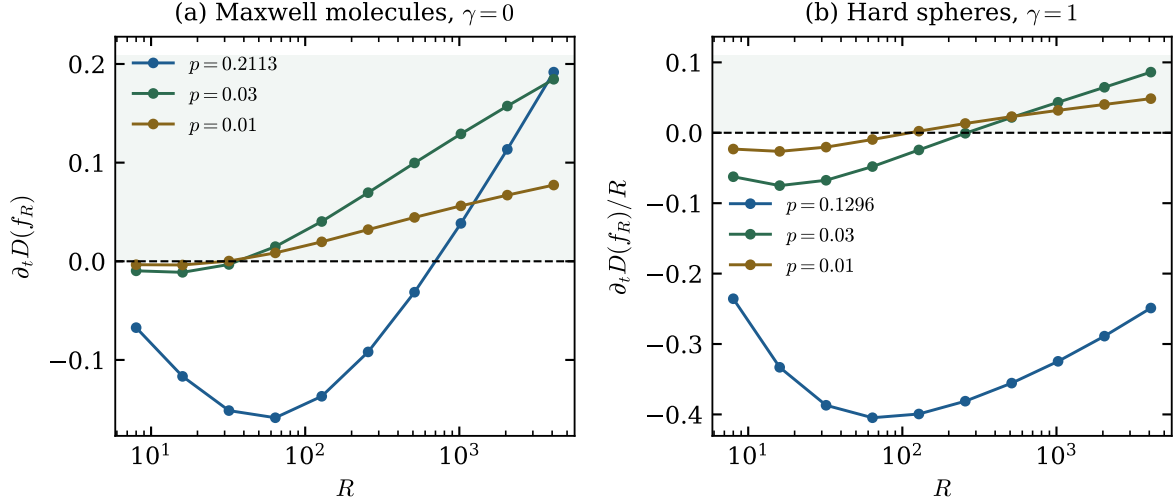


Figure 3: $\partial_t D(f_R)$ in Tier A test. Markers are our numerical values (converged up to 10^{-7} to 10^{-8}); the shaded half-plane is where McKean's conjecture fails. Note the vertical scale is $\partial_t D/R^\gamma$.

3 Tier B Test: time evolution of $\partial_t D(f)$ by 1D measure-valued algorithm (Maxwell molecules)

In this section, we implement Tier B test for time evolution of $\partial_t D(f)$: for $\gamma = 0$ only, Maxwellian mixtures are invariant under the flow, so the 3D equation reduces exactly to a closed 1D equation for the temperature-mixing measure, whose integration gives $D(f(t))$ itself and, as its initial slope, an independent measurement of the same $\partial_t D(f_R)$. More specifically, what we will essentially do in Tier B is to re-write [6, Lemma 4] into a 1D algorithm, evolving the temperature-mixing measure rather than a velocity-grid approximation.

3.1 Method

We can read (13) from the perspective of measures-valued solution: colliding M_a with M_b produces the uniform mixture of Maxwellians over the temperature interval between a and b , since $Q(f, f) = P(f, f) - f$ at $\gamma = 0$ for unit mass, and P is bilinear and symmetric.

Theorem 1 (exact 1D reduction). *For $\gamma = 0$ and $B = 1/(4\pi)$, and let $f(0, v) = \int \mu_0(T) M_T(v) dT$ be a mixture of Maxwellians of unit mass. Then,*

$$f(t, v) = \int M_T(v) \mu_t(dT), \quad \mu_0 = (1-p)\delta_1 + p\delta_R.$$

remains a mixture for all t , and the temperature-mixing measure satisfies the closed equation, i.e., the positive Maxwellian-mixture class is invariant under the Boltzmann flow, and its temperature measure satisfies

$$\partial_t \mu_t = \mathcal{B}[\mu_t, \mu_t] - \mu_t,$$

where

$$\mathcal{B}[\nu_1, \nu_2] = \iint \int_0^1 \delta_{(1-u)T+uT'} du \nu_1(dT) \nu_2(dT').$$

For $T \neq T'$, the inner integral is the uniform probability measure on $[\min(T, T'), \max(T, T')]$; for $T = T'$, it is δ_T . Thus the definition includes both mixed-temperature and equal-temperature collisions. It conserves $\int \mu$ (mass) and $\int T \mu$ (energy), and $\text{supp } \mu_t \subseteq [\min \text{supp } \mu_0, \max \text{supp } \mu_0]$.

The 3D Boltzmann equation therefore reduces *exactly* to a one-dimensional equation for μ . For the constructed f_R in [6] (corresponding to the case $\mu_0 = (1-p)\delta_1 + p\delta_R$) the atoms decouple exactly: writing $\mu_t = a(t)\delta_1 + b(t)\delta_R + g_t$, the diagonal of $\mathcal{B}$ on an atom is that atom again.

This equation preserves mass and the first temperature moment (hence kinetic energy), with $\text{supp } \mu_t \subset [1, R]$. It is an exact representation of the Boltzmann flow within this mixture class; numerical approximation enters only when the temperature measure and time are discretized.

For the initial state above, separate the endpoint atoms:

$$\mu_t = a(t)\delta_1 + b(t)\delta_R + g_t, \quad a(0) = 1-p, \quad b(0) = p, \quad g_0 = 0,$$

where g_t is the remaining nonnegative temperature measure and has no endpoint atoms.

By bilinearity and symmetry of $\mathcal{B}$,

$$\begin{aligned} \dot{a} &= a^2 - a, \\ \dot{b} &= b^2 - b, \\ \partial_t g_t &= 2ab U_{[1,R]} + 2aK_1[g_t] + 2bK_R[g_t] + \mathcal{B}[g_t, g_t] - g_t, \end{aligned}$$

with

$$U_{[1,R]}(dS) = \frac{\mathbf{1}_{[1,R]}(S)}{R-1} dS, \quad K_{T_0}[g] := \mathcal{B}[\delta_{T_0}, g].$$

Different temperature collisions produce continuous measures, so only equal-endpoint pairs contribute to the endpoint atoms. Consequently,

$$a(t) = \frac{a_0}{a_0 + (1-a_0)e^t}, \quad b(t) = \frac{b_0}{b_0 + (1-b_0)e^t}, \quad a_0 = 1-p, \quad b_0 = p.$$

The remaining measure g_t must still be evolved, after which the velocity density is reconstructed as

$$f(v, t) = a(t)M_1(v) + b(t)M_R(v) + \int M_T(v) g_t(dT).$$

At $t = 0$, this formulation gives

$$\dot{a}(0) = \dot{b}(0) = -p(1-p), \quad \partial_t g_t|_{t=0} = 2p(1-p)U_{[1,R]},$$

and therefore recovers the initial Boltzmann tangent

$$Q(f_R, f_R) = 2p(1-p) \left[\frac{1}{R-1} \int_1^R M_T dT - \frac{M_1 + M_R}{2} \right].$$

3.2 Numerical results

The identity behind the equivalence of (6) and (8) is $-2 \int Q_{\text{sym}}(f, Q) \log f = \int Q \mathcal{L}$. Tier B tests (6) end to end, by finite-differencing D along the *actual* flow and comparing against Tier A's evaluation of the static right-hand side. These are two computations with almost nothing in common: one integrates a nonlinear evolution equation for the temperature measure, the other evaluates a fourfold logarithmic collision integral.

R	p	Tier A, eq. (6)	$[D(h) - D(-h)]/2h$			Richardson	rel. diff.
			$h = 0.02$	$h = 0.01$	$h = 0.005$		
8	0.2	-0.06377808	-0.06377423	-0.06377332	-0.06377309	-0.06377302	7.9×10^{-5}
16	0.2	-0.10922677	-0.10922535	-0.10922033	-0.10921907	-0.10921865	7.4×10^{-5}
32	0.05	-0.01086482	-0.01086748	-0.01086070	-0.01085900	-0.01085844	5.9×10^{-4}
64	0.03	+0.01485566	+0.01483941	+0.01485568	+0.01485974	+0.01486110	3.7×10^{-4}
128	0.03	+0.04032973	+0.04026274	+0.04031821	+0.04033201	+0.04033662	1.7×10^{-4}

Table 4: Identity (6) verified against the actual dynamics, $\gamma = 0$. The last two rows are the Theorem 1: $\partial_t D > 0$, obtained by differentiating D along a solution of the Boltzmann equation.

Table 4 is, in our view, the strongest single piece of evidence in this report. It agrees to 10^{-4} – 10^{-5} relative, it simultaneously validates Tier A and the identity (6), and in its last two rows it exhibits $\partial_t D > 0$ as "an actual time derivative of the entropy production along a solution", not as the value of a functional.

Table 5 and Figure 4 are the picture the Theorem 1 describes: the entropy production rises before it falls. The $R = 16$, $p = 0.2$ curve is a control, and there D decreases monotonically, as McKean expected.

R	p	$D(f_R)$	$\max_t D$	at $t =$	rise
16	0.2	monotone decreasing (control case; $\partial_t D < 0$)			
128	0.03	0.231384	0.299380	2.20	+29.4%
1024	0.2113	1.664031	1.719341	0.57	+3.3%

Table 5: Tier B: the entropy production rises before it falls.

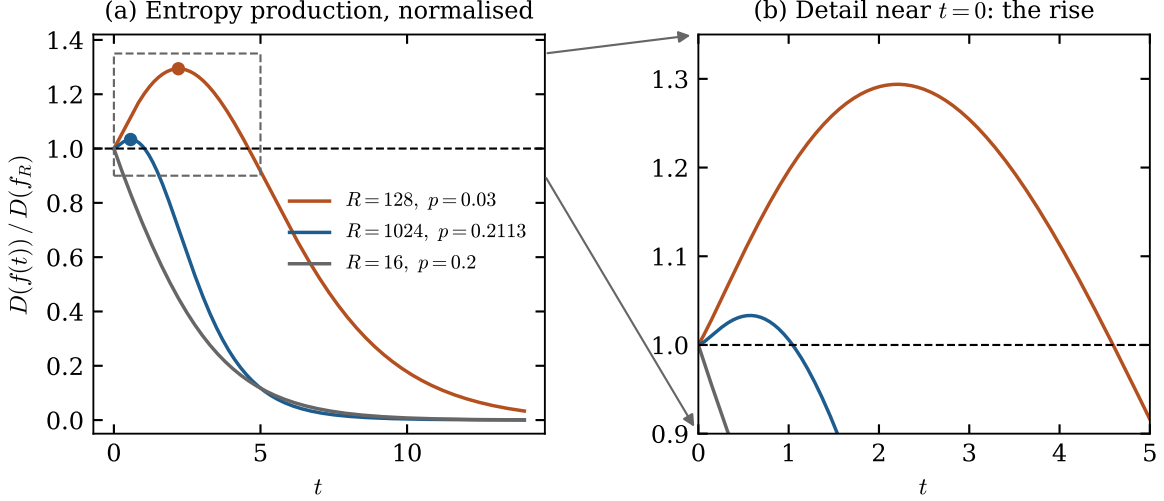


Figure 4: Tier B test: (a) Entropy production along the time evolution flow (Theorem 1). (b) magnifies the boxed window of (a) on the same normalized vertical axis.

4 Tier C Test: time evolution of $\partial_t D(f)$ by our 3D fast Fourier spectral solver (Maxwell molecules)

Tiers A and B both rely on the radial symmetry of the constructed solution. In this section, we present the Tier C test, which removes the symmetry the other two rely on: a 3D Cartesian Fourier spectral solver on a uniform velocity grid, assuming nothing about radial symmetry and taking Q from the discrete collision operator. The weights $1/f$ and $\mathcal{L}$ in (6) still come from the analytically known state, so Q is solved by our 3D Fourier spectral solver. Throughout, N is the number of points per direction on the periodic cube $D_L := [-L_{\text{box}}, L_{\text{box}}]^3$, so $k_{\text{max}} = \pi N / (2L_{\text{box}})$ is the largest resolved wavenumber; l and m are the two input Fourier modes and $k = l + m$ the output mode; R_{cut} truncates the relative speed $\rho = |v - v_*|$; and N_ρ and N_{sph} are the numbers of Gauss–Legendre nodes in ρ and of spherical-design directions in $\hat{q}$. S denotes the velocity-support radius of the datum.

4.1 Method

We apply our 3D fast Fourier spectral solver [2, 1] to approximate the collision operator Q : the Fourier spectral method seeks a solution $f \approx f_N \in \mathbb{P}_N$ such that, denoting $q = v - v_* = |q|\hat{q} = \rho\hat{q}$ and $c = \pi / (2L_{\text{box}})$,

$$f_N(t, v) = \sum_{k=-\frac{N}{2}}^{\frac{N}{2}-1} \hat{f}_k(t) e^{ick \cdot v}, \quad (16)$$

and substituting f_N into the Fourier transform of the collision operator, we have

$$\begin{aligned}
\hat{Q}_k &= \frac{1}{(2L_{\text{box}})^d} \int_{D_L} \int_{B_{R_{\text{cut}}}} \int_{\mathbb{S}^2} B f_N(v) f_N(v-q) (e^{-ick \cdot v'} - e^{-ick \cdot v}) d\sigma dq dv \\
&= \sum_{\substack{l, m = -\frac{N}{2} \\ l+m=k}}^{\frac{N}{2}-1} G(l, m) \hat{f}_l \hat{f}_m,
\end{aligned} \tag{17}$$

with the kernel mode $G(l, m)$ defined as

$$\begin{aligned}
G(l, m) &= \int_0^{R_{\text{cut}}} \int_{\mathbb{S}^2} |q|^2 e^{-ic|q|m \cdot \hat{q}} \left[\int_{\mathbb{S}^2} B(e^{ic|q|(l+m) \cdot (\hat{q}-\sigma)} - 1) d\sigma \right] d\hat{q} d|q| \\
&= \int_0^{R_{\text{cut}}} \int_{\mathbb{S}^2} |q|^2 e^{-ic|q|m \cdot \hat{q}} F(|q|, \hat{q}, l+m) d\hat{q} d|q|,
\end{aligned} \tag{18}$$

where the pre-computed $F(\rho, \hat{q}, k)$ is

$$F(\rho, \hat{q}, k) = \int_{\mathbb{S}^2} B(\rho, \hat{q} \cdot \sigma) \left[e^{ic\rho k \cdot (\hat{q}-\sigma)} - 1 \right] d\sigma.$$

In particular, for a VHS kernel with constant angular factor, $B = b_\gamma \rho^\gamma$, the σ integral is elementary — $\int_{\mathbb{S}^2} e^{-ia \cdot \sigma} d\sigma = 4\pi \text{sinc} |a|$ with $\text{sinc} x = \sin x/x$ — and

$$F(\rho, \hat{q}, k) = 4\pi b_\gamma \rho^\gamma \left[e^{ic\rho(k \cdot \hat{q})} \text{sinc}(c\rho|k|) - 1 \right]. \tag{19}$$

This is the reduction of following [5, Section 4.1]; we verified it against a high-accuracy product rule on the sphere to 10^{-13} . Since rebuilding a kernel mode is now a single complex exponential, caching F stops paying at all, and removing the cache is what lifts the memory ceiling. The half-shift phase separates across directions,

$$e^{ic\rho k \cdot \hat{q}} = \prod_{j=1}^3 e^{ic\rho k_j \hat{q}_j}, \tag{20}$$

an outer product of three one-dimensional exponentials: $3N$ complex exponentials per $(\rho, \hat{q})$ node instead of N^3 . Finally, $\text{sinc}(0) = 1$ is set as exact, so $F(\rho, \hat{q}, 0) = 0$ and mass is conserved upto machine precision whatever the ρ and $\hat{q}$ quadratures do.

Remark 1 (N_{sph} , not N , is the binding parameter). From our experience when doing test, once (19) has removed the σ quadrature, the $\hat{q}$ integrand still oscillates on the same $c\rho|k|$ scale, and the design rule for that quadrature limits everything. In the static benchmark — $Q(f_R, f_R)$ against the exact $2p(1-p)A_R$ at $R = 4$, $p = 0.2$, $N_\rho = 40$, $\gamma = 0$ — the relative error stalls near 20% at $N_{\text{sph}} = 12$ and never improves with N , whereas $N_{\text{sph}} = 192$ reaches 2.9×10^{-5} at $N = 64$. Since the cost is $O(N_\rho N_{\text{sph}} N^3 \log N)$, budget N_{sph} first.

Remark 2 (Use (6) instead of (8)). The seemingly more natural form (8) needs only the collision operator, Q_{sym} being the exact of the quadratic form the solver already computes. Over the whole box it fails outright: at $(\gamma, p, R) = (0, 0.03, 64)$, with $N = 240$, $N_{\text{sph}} = 192$ and an operator accurate to 1.3×10^{-3} , we obtained $\partial_t D = -0.203$ against the reference $+0.014856$ — the wrong sign, by a factor of 14. The cause is spectral tail noise, not interior accuracy: the error in Q is spread over the box at roughly its global level while the exact Q decays like a Gaussian, and both weights in (8) grow outward.

Silvestre’s identity (6) avoids both defects. There Q enters the second term *linearly*, with no lossy difference, and the weight $\mathcal{L}$ grows only like $|v|^\gamma$ — bounded for $\gamma = 0$, where it saturates near 7.9, and linear in $|v|$ for hard spheres — instead of growing quadratically. Since $\mathcal{L}$ and f_R are properties of the analytically known datum, taking them from Tier A is exactly as legitimate as taking $\log f$ and $1/f$ analytically: the only spectral input remains Q . The $1/f$ term still needs a radial cutoff, but truncation is not an error if the reference is truncated identically — so we compare against Tier A’s *truncated* integrals at the same r_{cut} , at one collision evaluation per case instead of three. The truncated $\partial_t D$ is negative for $r_{\text{cut}} \lesssim \sqrt{R}$ and positive beyond, so its sign structure is itself a test, and truncation *improves* the conditioning: at $(1, 0.005, 128)$ the cancellation falls from $9.1\times$ to $1.7\times$ at $r_{\text{cut}} = 2\sqrt{R}$. The geometric mechanism behind this is illustrated in Figure 5.

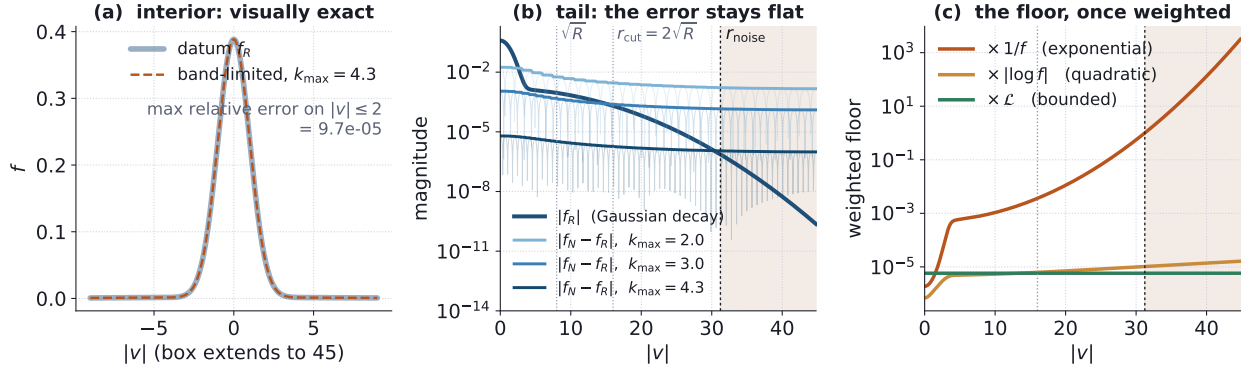


Figure 5: The absolute error floor, on a one-dimensional model of f_R at $(\gamma, p, R) = (0, 0.03, 64)$ to illustrate the mechanism behind Remark 2.

4.2 Numerical results

In practical implementation, the trajectory comes from the invariant class, not from the grid. For $\gamma = 0$, [6, Lemma 4] makes the Maxwellian mixtures invariant, so Tier B supplies

$$f(t) = a(t)M_1 + b(t)M_R + \sum_i m_i(t)M_{T_i}, \quad a, b, m_i \geq 0, \quad (21)$$

an exact-in-class, strictly positive, radial state at every t , together with its exact $Q(f, f) = \partial_t f$. Each diagnostic time is then the same instantaneous evaluation that Table 6 reports, at a state other than f_R : f and Q_{ex} are built on the 3D grid directly from (21), summed over its ≈ 1200 terms with no interpolation, so $1/f$ is exactly as clean as at $t = 0$; $\mathcal{L}$ comes from the Tier A radial quadrature of the same mixture; and Q remains the one spectral solver input.

$r_{\text{cut}}/\sqrt{R}$	$\partial_t D$ spectral	$\partial_t D$ Tier A	rel. err.	sign
$(\gamma, p, R) = (0, 0.03, 64)$, <i>Maxwell molecules</i> : $N = 128$, $S = 28.00$, $L_{\text{box}} = 44.80$, $k_{\text{max}} = 4.49$, <i>rel. err. of $Q = 7.0 \times 10^{-4}$</i>				
1.0	-0.004816	-0.004337	1.1×10^{-1}	✓
1.5	+0.032754	+0.032824	2.1×10^{-3}	✓
2.0	+0.036103	+0.038138	5.3×10^{-2}	✓
$(\gamma, p, R) = (0, 0.003, 48)$, <i>Maxwell molecules</i> : $N = 128$, $S = 24.25$, $L_{\text{box}} = 38.80$, $k_{\text{max}} = 5.18$, <i>rel. err. of $Q = 5.6 \times 10^{-4}$</i>				
1.0	-0.000038	+0.000002	1.8×10^1	×
1.5	+0.004016	+0.004022	1.5×10^{-3}	✓
2.0	+0.004598	+0.004689	1.9×10^{-2}	✓
$(\gamma, p, R) = (0, 0.2, 16)$, <i>Maxwell molecules</i> : $N = 128$, $S = 14.00$, $L_{\text{box}} = 22.40$, $k_{\text{max}} = 8.98$, <i>rel. err. of $Q = 3.1 \times 10^{-4}$</i>				
1.0	-0.084335	-0.084970	7.5×10^{-3}	✓
1.5	-0.074284	-0.074125	2.1×10^{-3}	✓
2.0	-0.061641	-0.061610	5.0×10^{-4}	✓
$(\gamma, p, R) = (1, 0.002, 96)$, <i>hard spheres</i> : $N = 128$, $S = 34.29$, $L_{\text{box}} = 54.87$, $k_{\text{max}} = 3.66$, <i>rel. err. of $Q = 2.5 \times 10^{-1}$</i>				
1.0	-0.312729	-0.049265	5.3×10^0	✓
1.5	+0.423974	+0.935161	5.5×10^{-1}	✓
2.0	+0.690182	+1.424528	5.2×10^{-1}	✓
$(\gamma, p, R) = (1, 0.002, 112)$, <i>hard spheres</i> : $N = 128$, $S = 37.04$, $L_{\text{box}} = 59.26$, $k_{\text{max}} = 3.39$, <i>rel. err. of $Q = 6.4 \times 10^{-1}$</i>				
1.0	-1.054328	-0.023827	4.3×10^1	✓
1.5	-0.528225	+1.195710	1.4×10^0	×
2.0	-0.554671	+1.784995	1.3×10^0	×
$(\gamma, p, R) = (1, 0.002, 96)$, <i>hard spheres</i> : $N = 128$, $S = 29.39$, $L_{\text{box}} = 47.03$, $k_{\text{max}} = 4.28$, <i>rel. err. of $Q = 1.5 \times 10^{-2}$</i>				
1.0	-0.061208	-0.049265	2.4×10^{-1}	✓
1.5	+0.922799	+0.935161	1.3×10^{-2}	✓
2.0	+1.261217	+1.424528	1.1×10^{-1}	✓

Table 6: Tier C: $\partial_t D(f_R)$ from $-\int_{|v| \leq r_{\text{cut}}} Q_{\text{spec}}^2 / f_R + \int_{|v| \leq r_{\text{cut}}} Q_{\text{spec}} \mathcal{L}$ against Tier A truncated identically, with $L_{\text{box}} = 1.6S$ throughout. Bold rows are those where the reference is positive, i.e. where McKean’s conjecture fails.

Table 7 and Figure 6 present the result with most accurate positive case: $(\gamma, p, R) = (0, 0.03, 64)$ at $N = 128$, $S = 3.5\sqrt{R}$, $L_{\text{box}} = 1.6S$, $N_\rho = 25$, $N_{\text{sph}} = 192$. The $t = 0$ row reproduces Table 6, which is the pipeline’s own check. The two reference columns are computed by different quadratures and differ in the sixth decimal (+0.032824 against +0.032825, 1.0×10^{-5} relative), two orders below the spectral error.

t	rel. err. of Q	$r_{\text{cut}} = 1.5\sqrt{R}$			$r_{\text{cut}} = 2\sqrt{R}$		
		Tier C	Tier A	rel. err.	Tier C	Tier A	rel. err.
0.00	7.0×10^{-4}	+0.032754	+0.032825	2.2×10^{-3}	+0.036103	+0.038138	5.3×10^{-2}
0.25	5.8×10^{-4}	+0.037091	+0.037131	1.1×10^{-3}	+0.036915	+0.038904	5.1×10^{-2}
0.50	4.9×10^{-4}	+0.035766	+0.035793	7.5×10^{-4}	+0.032988	+0.034918	5.5×10^{-2}
0.75	4.1×10^{-4}	+0.032368	+0.032385	5.3×10^{-4}	+0.027672	+0.029521	6.3×10^{-2}
1.00	3.6×10^{-4}	+0.027980	+0.027989	3.2×10^{-4}	+0.021896	+0.023648	7.4×10^{-2}
1.25	3.2×10^{-4}	+0.023098	+0.023100	1.0×10^{-4}	+0.016048	+0.017688	9.3×10^{-2}
1.50	2.8×10^{-4}	+0.018012	+0.018009	1.6×10^{-4}	+0.010329	+0.011850	1.3×10^{-1}
1.75	2.5×10^{-4}	+0.012916	+0.012909	5.4×10^{-4}	+0.004866	+0.006262	2.2×10^{-1}
2.00	2.3×10^{-4}	+0.007950	+0.007940	1.3×10^{-3}	-0.000260	+0.001010	1.3×10^0
2.50	2.0×10^{-4}	-0.001205	-0.001221	1.3×10^{-2}	-0.009287	-0.008265	1.2×10^{-1}
3.00	1.8×10^{-4}	-0.008912	-0.008933	2.3×10^{-3}	-0.016488	-0.015693	5.1×10^{-2}

Table 7: Tier C: $\partial_t D(f(t))$ from the 3D spectral operator against Tier A, along the exact Maxwellian-mixture trajectory through f_R at $(\gamma, p, R) = (0, 0.03, 64)$, $N = 128$, $S = 3.5\sqrt{R}$, $L_{\text{box}} = 1.6S$, $k_{\text{max}} = 4.49$.

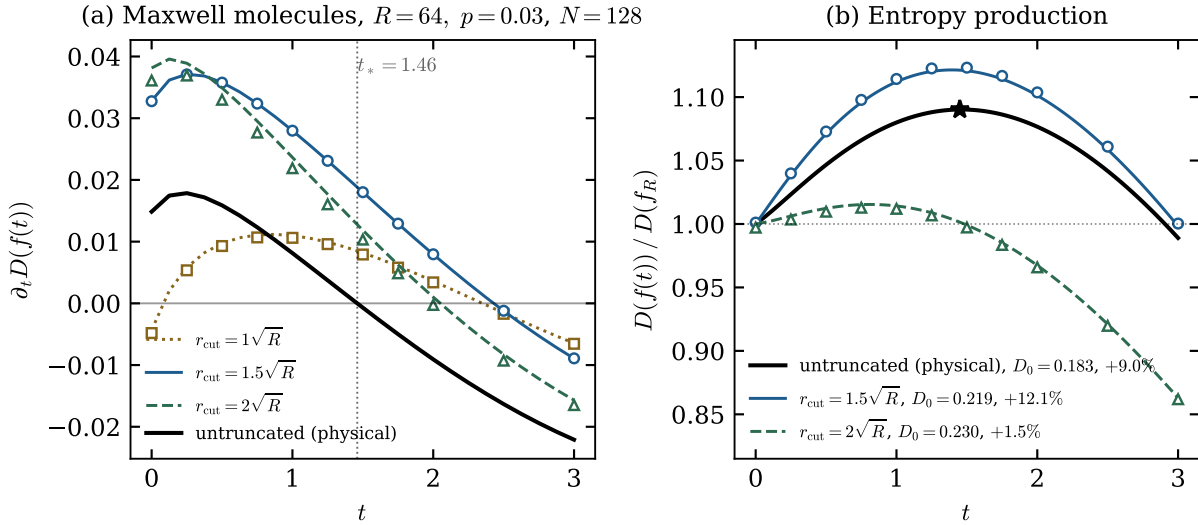


Figure 6: Tier C at $(\gamma, p, R) = (0, 0.03, 64)$, $N = 128$. (a) $\partial_t D(f(t))$ for the three truncation windows and the untruncated integral; lines are the Tier A radial reference on the same states, markers the 3D spectral operator, one collision evaluation each. (b) the entropy production $D(f(t))$ itself, rising to a maximum and falling; the star marks the physical maximum, at $t = 1.45$.

Cost. One collision evaluation at $N = 128$, $N_\rho = 25$, $N_{\text{sph}} = 192$ took 226–334 s over the seventeen evaluations reported here, and Tiers A and B are minutes of Python.

5 Summary and outlook

We summarize what we did in this report go beyond numerically verifying the results of [6].

- Tier A: The γ -extension of [6, Lemma 4] (i.e., Proposition 1) gives the two Maxwellian gain term in closed enough form to evaluate for any $\gamma \in [0, 1]$.

- Tier B: The exact reduction of Theorem 1 — Maxwellian mixtures are invariant under the $\gamma = 0$ flow, with a closed 1D equation for the mixing measure, which makes the dynamic computation cheap at any R .
- Tier C: The 3D spectral solver is accurate enough to reproduce the sign change of $\partial_t D$ along a trajectory, and can possibly be used to explore the non-radial or non-cutoff solution that may fail McKean’s conjecture in the future.

Disclosure

The algorithms and code used to produce the results in this report are available at <https://github.com/kunlun-qi/Numerical-disproof-McKean-conjecture.git>. We use Claude Opus 5 to assist in the coding of the numerical simulations for the Tier A and Tier B test, whereas Tier C is based on our previous solver in [2].

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